SECSR2043 OPERATING SYSTEMS
[20 Marks]

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Section	:	02	

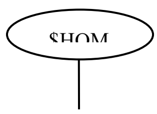
Instruction: Please answer all of the following questions. Whenever the ☝ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo “Hello world” > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo “Honda Accord” > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **$HOME** directory.

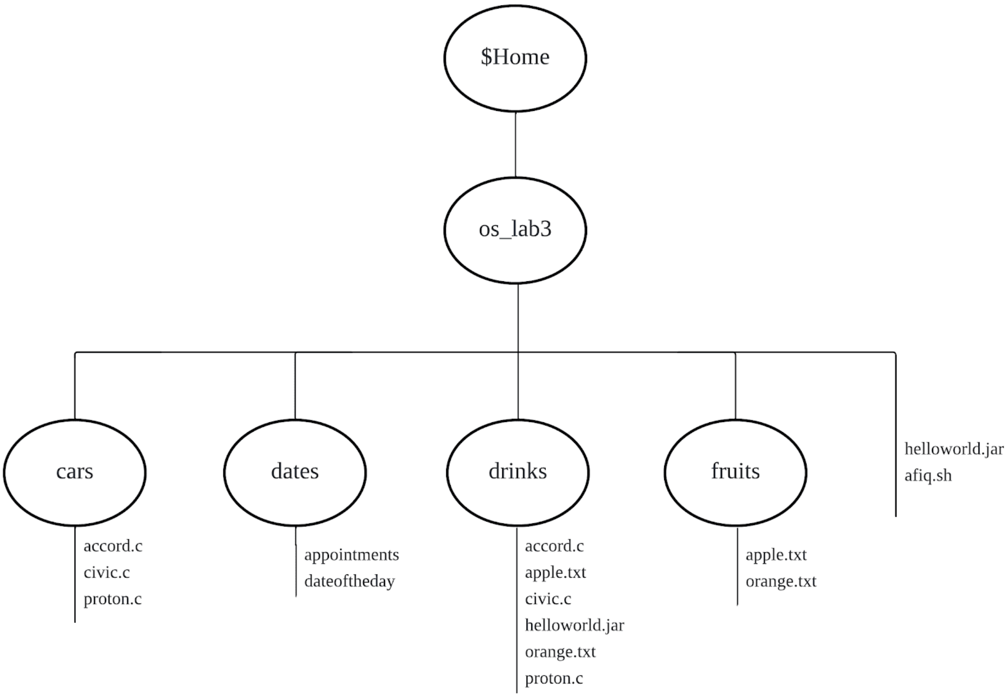
[4 marks]



Print screen the script that you type:

```
GNU nano 7.2
echo  "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c;
cd ../dates; date > dateoftheday cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

Then draw the tree



b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
  GNU nano 7.2                          afiq1b.sh *
#!/bin/bash

read -p "Enter the file extension (without the dot): " ext

echo "Files with .${ext} extension:"
find -type f -name "*.${ext}"

count=$(find -type f -name "*.${ext}" | wc -l)
echo "Number of files with .${ext} extension: $count"
```

```
afiq@secr2043:~/os_lab3$ chmod u+x afiq1b.sh
afiq@secr2043:~/os_lab3$ ./afiq1b.sh
Enter the file extension (without the dot): c
Files with .c extension:
./fruits/civic.c
./fruits/accord.c
./fruits/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Number of files with .c extension: 6
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

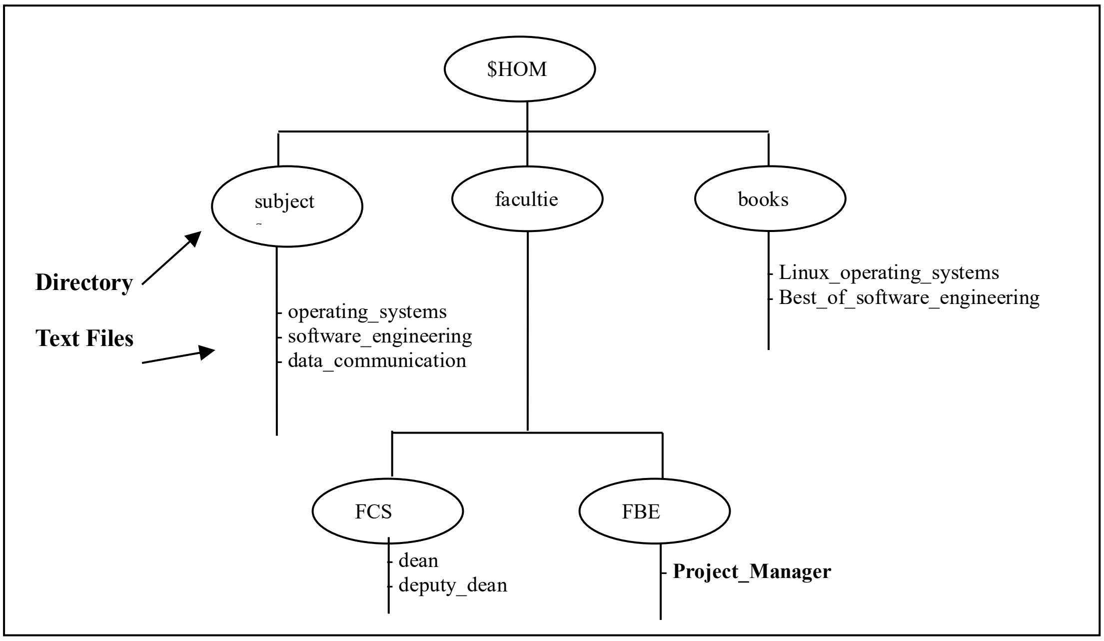


Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2                                afiq2a.sh
#!/bin/bash

mkdir -p $HOME/subject $HOME/faculti/FCS $HOME/faculti/FBE $HOME/books

echo "operating systems" > $HOME/subject/operating_systems
echo "software engineering" > $HOME/subject/software_engineering
echo "data communication" > $HOME/subject/data_communication

echo "dean" > $HOME/faculti/FCS/dean
echo "deputy dean" > $HOME/faculti/FCS/deputy_dean

echo "Project Manager" > $HOME/faculti/FBE/Project_Manager

echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering
```

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`. [2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subject	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--

```
drwxrwxr-x 2 afiq afiq 4096 Jun 28 09:10 books
drwxrwxr-x 2 afiq afiq 4096 Jun 28 09:10 subject
-rw-rw-r-- 1 afiq afiq 29 Jun 28 09:10 Best_of_software_engineering
drwxrwxr-x 2 afiq afiq 4096 Jun 28 09:10 FCS
-rw-rw-r-- 1 afiq afiq 16 Jun 28 09:10 Project_Manager
```

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
`Subject`	✓	✓	✓	✓	x	x	✓	x	x
`Best_of_software_engineering`	✓	x	✓	x	✓	x	x	x	x
`FCS`	✓	✓	x	x	x	x	✓	✓	✓
`project_manager`	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
  GNU nano 7.2                              afiq2c.sh
#!/bin/bash

chmod 744 $HOME/subject

chmod 520 $HOME/books/Best_of_the_engineering

chmod 607 $HOME/faculti/FCS

chmod 031 $HOME/faculti/FBE/Project_Manager
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
`subject`	drwxr--r--
`Best_of_software_engineering`	-r-x-w----
`FCS`	drw----rwx
`project_manager`	-----wx--x

```
drwxr--r-- 2 afiq afiq 4096 Jun 28 09:10 subject
-r-x-w---- 1 afiq afiq 29 Jun 28 09:10 Best_of_software_engineering
drw----rwx 2 afiq afiq 4096 Jun 28 09:10 FCS
-----wx--x 1 afiq afiq 16 Jun 28 09:10 Project_Manager
```

End of Lab 3

**** All the Best for Final Exam ****